A particle travels in a line with velocity

$$v(t) = 2 - t$$

What is its displacement between $t = 0$ and $t = 5$?

$$s(5) - s(0) = \int_0^5 v(t)\,dt$$

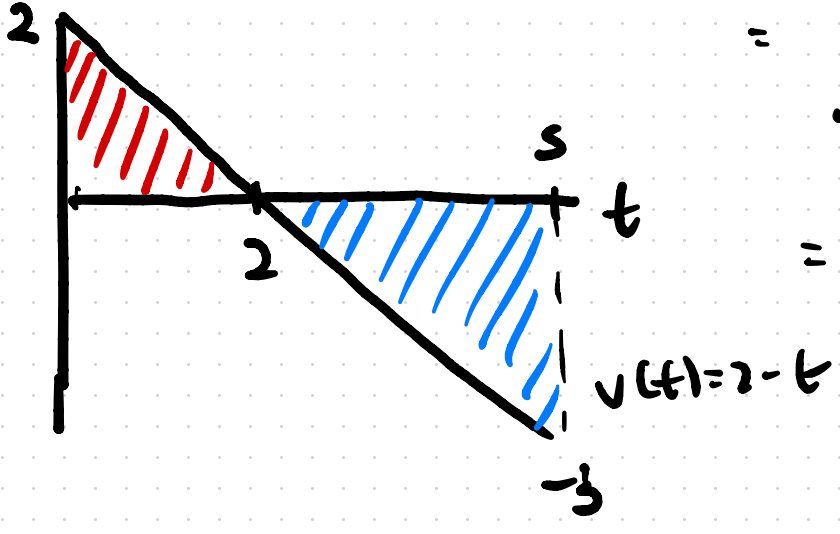
$$= \int_0^5 (2 - t)\,dt$$

$$= 2t - \frac{1}{2}t^2 \Big|_0^5$$

$$= -\frac{5}{2}$$

What is the distance traveled between $t=0$ and $t=5$?

$$\int_0^5 |v(t)|\,dt = \int_0^5 |2-t|\,dt$$



$$= \int_0^2 (2-t)\,dt + \int_2^5 (t-2)\,dt$$

$$= \frac{1}{2}\cdot 2\cdot 2 + \frac{1}{2}\cdot 3\cdot 3$$

area

$$= \frac{13}{2}$$

The Substitution Rule (5.5)

$$\int e^{x^2}\,dx$$

No elementary form

$\int_0^x e^{t^2}\,dt$ is an antiderivative, but this is as simple as it gets.

$$\int x e^{x^2}\,dx$$

chain rule

$$\frac{d}{dx} e^{x^2} = 2x e^{x^2}$$

$$\frac{d}{dx} \frac{1}{2} e^{x^2} = x e^{x^2}$$

$$\boxed{\frac{1}{2} e^{x^2} + C}$$

Recall the chain rule:

$$(f(g(x))' = f'(g(x))\,g'(x).$$

In other words, $f(g(x))$ is an antiderivative of $f'(g(x))\,g'(x)$. Thus,

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C$$

In practice, it is often hard to recognize functions of the form $f'(g(x))\,g'(x)$.

Substitution rule: (u-sub)

Let $u = g(x)$. Then

"differential of u" $du = g'(x)dx$

$\frac{du}{dx} = g'(x)$

$du = g'(x)dx$

and we can replace any instance of $g'(x)dx$ in an integral with du.

$$\int x^2 \sin(x^3+1)\,dx$$

$u = x^3+1$

$du = 3x^2\,dx$

$\frac{1}{3}du = x^2\,dx$

$$\int \sin(x^3+1)\; x^2\,dx$$

$$\int \sin u\; \frac{1}{3}du$$

$$-\frac{1}{3}\cos u + C$$

$$-\frac{1}{3}\cos(x^3+1) + C$$

$$\int \frac{x}{\sqrt{1+x^2}}\,dx$$

$u = 1+x^2$

$du = 2x\,dx$

$\frac{1}{2}du = x\,dx$

$$\int \frac{1}{\sqrt{u}}\,\frac{1}{2}\,du$$

$$\sqrt{u} + C$$

$$\sqrt{1+x^2} + C$$

$$\int \frac{x^3}{\sqrt{1+x^2}}\,dx \qquad x^2 \cdot x$$

$u = 1 + x^2$

$du = 2x\,dx$

$\frac{1}{2}du = x\,dx$

$x^2 = u - 1$

$$\int \frac{x^2}{\sqrt{u}} \cdot \frac{1}{2}\,du$$

$$\int \frac{u-1}{\sqrt{u}} \frac{1}{2}\,du$$

$$\int \frac{1}{2}\left(\sqrt{u} - \frac{1}{\sqrt{u}}\right)du$$

$$\frac{1}{2}\left(\frac{2}{3}u^{3/2} - 2u^{1/2}\right) + C$$

$$\frac{1}{3}u^{3/2} - u^{1/2} + C$$

$$\frac{1}{3}(1+x^2)^{3/2} - (1+x^2)^{1/2} + C$$

$$\int \frac{\ln x}{x}\,dx$$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$$\int \ln x \left(\frac{1}{x}\,dx\right)$$

$$\int u\,du$$

$$\frac{1}{2}u^2 + C$$

$$\frac{1}{2}(\ln x)^2 + C$$

$$\int_0^1 x\sqrt{2x^2+1}\,dx$$

Method 1: Find an antiderivative $F(x)$ of $x\sqrt{2x^2+1}$, and then evaluate $F(1)-F(0)$.

Method 2: Do the substitution rule "in place".

$u = 2x^2+1$

$du = 4x\,dx$

$\frac{1}{4}du = x\,dx$

need to change bounds to u

$$\int_0^1 x\sqrt{2x^2+1}\,dx$$

$$u = 2x^2+1$$

@ $x=0$ $\quad u=1$

@ $x=1$ $\quad u=3$

$$\int_1^3 \sqrt{u}\,\frac{1}{4}\,du$$

bounds for u

$$\frac{1}{4}\cdot\frac{2}{3}u^{3/2}\Big|_{u=1}^{u=3}$$

$$\frac{1}{6}\cdot 3^{3/2} - \frac{1}{6}$$

$$\int_0^2 x\sqrt{3-x}\,dx$$

$u = 3 - x$

$du = -dx$

$-du = dx$

$x = 3 - u$

@ $x = 0$ $u = 3$

@ $x = 2$ $u = 1$

$$\int_3^1 x\sqrt{u}\,(-du)$$

$$\int_3^1 (3-u)\sqrt{u}\,(-du)$$

$$\int_3^1 (u-3)\sqrt{u}\,du$$

$$\int_3^1 \left(u^{3/2} - 3u^{1/2}\right)du$$

$$\frac{2}{5}u^{5/2} - \cancel{2}\cdot\frac{2}{\cancel{2}}u^{3/2}\Big|_3^1$$

$$\left(\frac{2}{5} - 2\right) - \left(\frac{2}{5}\cdot 3^{5/2} - 2\cdot 3^{3/2}\right)$$

$$-\frac{2}{5}\cdot 3^{5/2} + 2\cdot 3^{3/2} - \frac{8}{5}$$